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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 2

$$f(x) = (x - 2)^3 + 1$$

Nulpunten:

$$f(x) = 0 \Rightarrow f(1) = (1 - 2)^3 + 1 = 0 \Rightarrow x = 1$$

$$f'(x) = 3(x - 2)^2$$

Extrema:

$$f'(x) = 0 \Rightarrow f(2) = 3(2 - 2)^2 = 0 \Rightarrow x = 2$$

$$f''(x) = 6(x - 2)$$

Buigpunten:

$$f''(x) = 0 \Rightarrow f''(2) = 3(2 - 2)^2 = 0 \Rightarrow x = 2$$

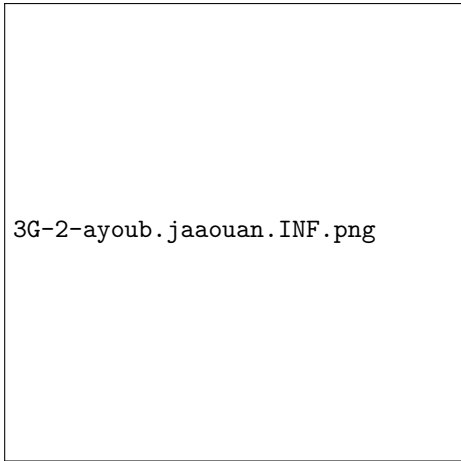
Tekenonderzoek.:

x	$-\infty$	1		2	$-\infty$
$f(x)$	—	0	+	+	+
$f'(x)$	+	+	+	0 ⁽²⁾	+
$f''(x)$	—	—	—	0	+
$f(x)$	$\frown$	0	$\frown$	B	$\smile$
	$\nearrow$	0	$\nearrow$	(1)	$\nearrow$

Raaklijn in punt $(1, 0)$:

$$y = 3(x - 1)$$

References



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